Grade 7
Unit 9 Assessment

Name____________________
Date_________________
Period__________

1. Find the sum of each expression using the fewest terms possible.

a. $(x + 9) + (2x + 3)$

$3x + 12$

b. $(4y - 7) + (16 + 6y)$

$10y + 9$

2. Find the difference of each expression using the fewest terms possible.

a. $(11k + 15) - (7k + 8)$

$4k + 7$

b. $(8 - 2x) - (-5x + 4)$

$3x + 4$

3. Complete each statement.

a. The

- associative
- commutative
- distributive

property can be used to state that $(d+4)8+7d$ is equivalent to $8d+32+7d$.

b. The

- associative
- commutative
- distributive

property can be used to state that $\frac{6}{5}x+\left(\frac{3}{4}x+\frac{4}{5}\right)$ is equivalent to $\left(\frac{3}{4}x+\frac{4}{5}\right)+\frac{6}{5}x$.

4. Write each expression using the fewest possible terms.

a. $14.8-(6.3-2.9r)$	b. $(5.9x-1.7-3.76x)+(14.78+16.3x)$
$2.9r+8.5$	$18.44x+13.08$

5. What is the result when $\frac{7}{8}w+\frac{5}{6}$ is subtracted from $\frac{14}{4}w-\frac{23}{6}$?

$$\frac{21}{8}w-\frac{28}{6}$$

6. What is the result when $8\frac{9}{20}x - 7\frac{3}{5}$ is added to $11\frac{1}{5} - 9\frac{3}{4}x$?

$$-1\frac{3}{20}x + 3\frac{13}{20}$$

7. Use the given expressions to answer each.

$14(2m + 5) - 2$ and $28m + 3$

a. Choose a value for m to determine if the expressions are equivalent. Show your work.

Answers will vary. Use of $m = 0$ will show equivalence and is the only value that should not be used.

b. Use properties of operations to determine if the expressions are equivalent. Show your work.

$$28m + 70 - 2$$

$$28m + 68$$

c. Complete the statement.

The expressions
- ○ are
- ○ are not

equivalent.

8. Write an equivalent expression only using the property of operations given.

Expression	Property	Equivalent Expression
$(5 - 0.7c) + (1.2c - 4.8)$	**Commutative**	$(1.2c - 4.8) + (5 - 0.7c)$ **or** $(1.2c - 4.8) + (-0.7c + 5)$ **or** $(-4.8 + 1.2c) + (5 - 0.7c)$
$12y + (10y + 4) - 17$	**Associative**	$(12y + 10y) + 4 - 17$ **or** $12y + 10y + (4 - 17)$
$11x - 7(6 - 9x) + 23$	**Distributive**	$11x - 42 + 63x + 23$